```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go


%matplotlib inline

import warnings
warnings.filterwarnings('ignore')
```

```
superstore_sales = pd.read_excel('US Superstore data.xls')
superstore_sales.sample(8)
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Postal Code	Region	Product ID
7948	7949	CA-2014-131009	2014-03-01	2014-03-05	Standard Class	SC-20380	Shahid Collister	Consumer	United States	El Paso	...	79907	Central	FUR-CH-10001270
2587	2588	CA-2017-131695	2017-06-30	2017-07-06	Standard Class	RA-19285	Ralph Arnett	Consumer	United States	New York City	...	10024	East	FUR-FU-10002045
2421	2422	CA-2016-155551	2016-04-18	2016-04-23	Standard Class	CR-12580	Clay Rozendal	Home Office	United States	Elmhurst	...	60126	Central	OFF-ST-10003656
2207	2208	US-2014-103905	2014-07-14	2014-07-20	Standard Class	AW-10930	Arthur Wiediger	Home Office	United States	Aurora	...	60505	Central	TEC-PH-10001552
2894	2895	CA-2017-139913	2017-10-23	2017-10-29	Standard Class	JC-16105	Julie Creighton	Corporate	United States	Rochester	...	14609	East	FUR-FU-10000771
1997	1998	CA-2014-131905	2014-02-06	2014-02-09	First Class	ND-18460	Neil Ducich	Corporate	United States	Chesapeake	...	23320	South	OFF-LA-10002787
7935	7936	CA-2017-166093	2017-08-17	2017-08-24	Standard Class	RW-19540	Rick Wilson	Corporate	United States	Brentwood	...	94513	West	OFF-PA-10003893
5636	5637	CA-2017-123022	2017-09-03	2017-09-08	Standard Class	SC-20050	Sample Company A	Home Office	United States	La Mesa	...	91941	West	TEC-AC-10000474

8 rows × 21 columns

EXPLORATORY DATA

```
df = superstore_sales.copy()
```

```
df.shape
```

```
(9994, 21)
```

```
df.columns
```

```
Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
       'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
       'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',
       'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   datetime64[ns]
 3   Ship Date      9994 non-null   datetime64[ns]
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: datetime64[ns](2), float64(3), int64(3), object(13)
memory usage: 1.6+ MB
```

```
df[['Sales','Discount','Profit','Quantity']].describe()
```

	Sales	Discount	Profit	Quantity
count	9994.000000	9994.000000	9994.000000	9994.000000
mean	229.858001	0.156203	28.656896	3.789574
std	623.245101	0.206452	234.260108	2.225110
min	0.444000	0.000000	-6599.978000	1.000000
25%	17.280000	0.000000	1.728750	2.000000
50%	54.490000	0.200000	8.666500	3.000000
75%	209.940000	0.200000	29.364000	5.000000
max	22638.480000	0.800000	8399.976000	14.000000

```
df=df.drop('Row ID',axis=1)
```

```
df['Country'].value_counts()
```

Country	count
United States	9994

dtype: int64

```
df=df.drop('Country',axis=1)
```

```
df.drop(['Order ID',         'Ship Date',          'Ship Mode',          'Customer ID',          'Customer Name',
```

```
df.shape
```

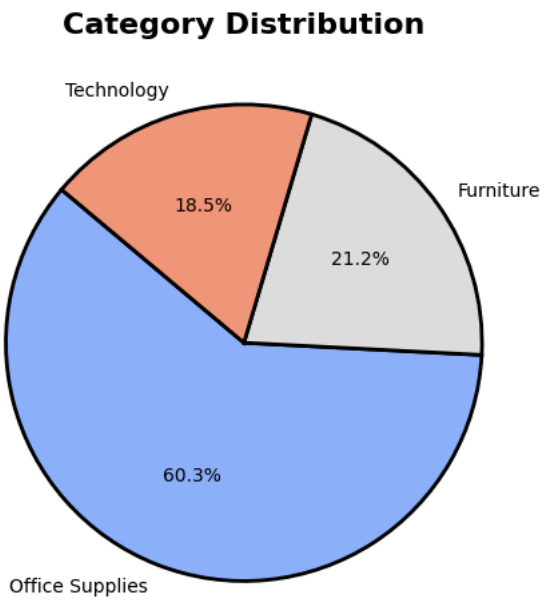
```
(9994, 11)
```

```
category_counts = df['Category'].value_counts()
category_counts
```

Category	count
Office Supplies	6026
Furniture	2121
Technology	1847

dtype: int64

```
fig, axs = plt.subplots(figsize=(5, 5))

axs.pie(
    category_counts,
    labels=category_counts.index,
    autopct='%1.1f%%',
    startangle=140,
    colors=sns.color_palette('coolwarm', len(category_counts)),
    wedgeprops={'edgecolor': 'black', 'linewidth': 2}
)

axs.set_title('Category Distribution', fontsize=16, fontweight='bold')

plt.tight_layout()
plt.show()
```

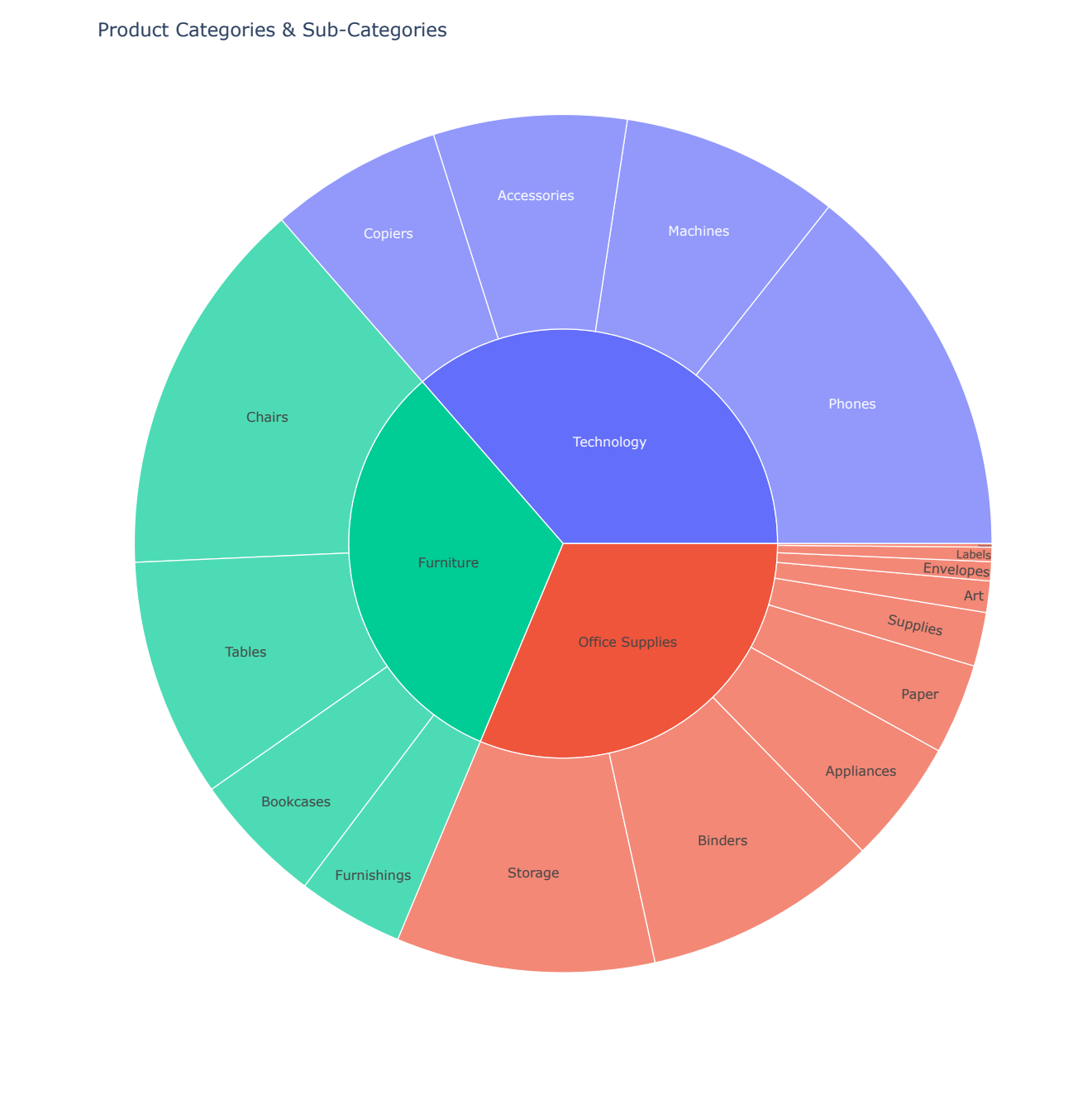


Category Distribution (Pie Chart)

The pie chart shows how your data is distributed across different product categories. One or two categories likely dominate the dataset, contributing the largest share. Smaller slices indicate less frequent categories or lower representation. This helps quickly identify which category is most common or important in your dataset.

Insight: If one category is significantly larger, your business/data is heavily dependent on that segment. Balanced slices indicate diversification across categories.

```
import plotly.express as px

fig = px.sunburst(
    df,
    path=['Category', 'Sub-Category'],
    values='Sales',
    color='Category',
    hover_data=['Sales', 'Quantity', 'Profit']
)

fig.update_layout(
    height=1000,
    title_text='Product Categories & Sub-Categories'
)

fig.show()
```

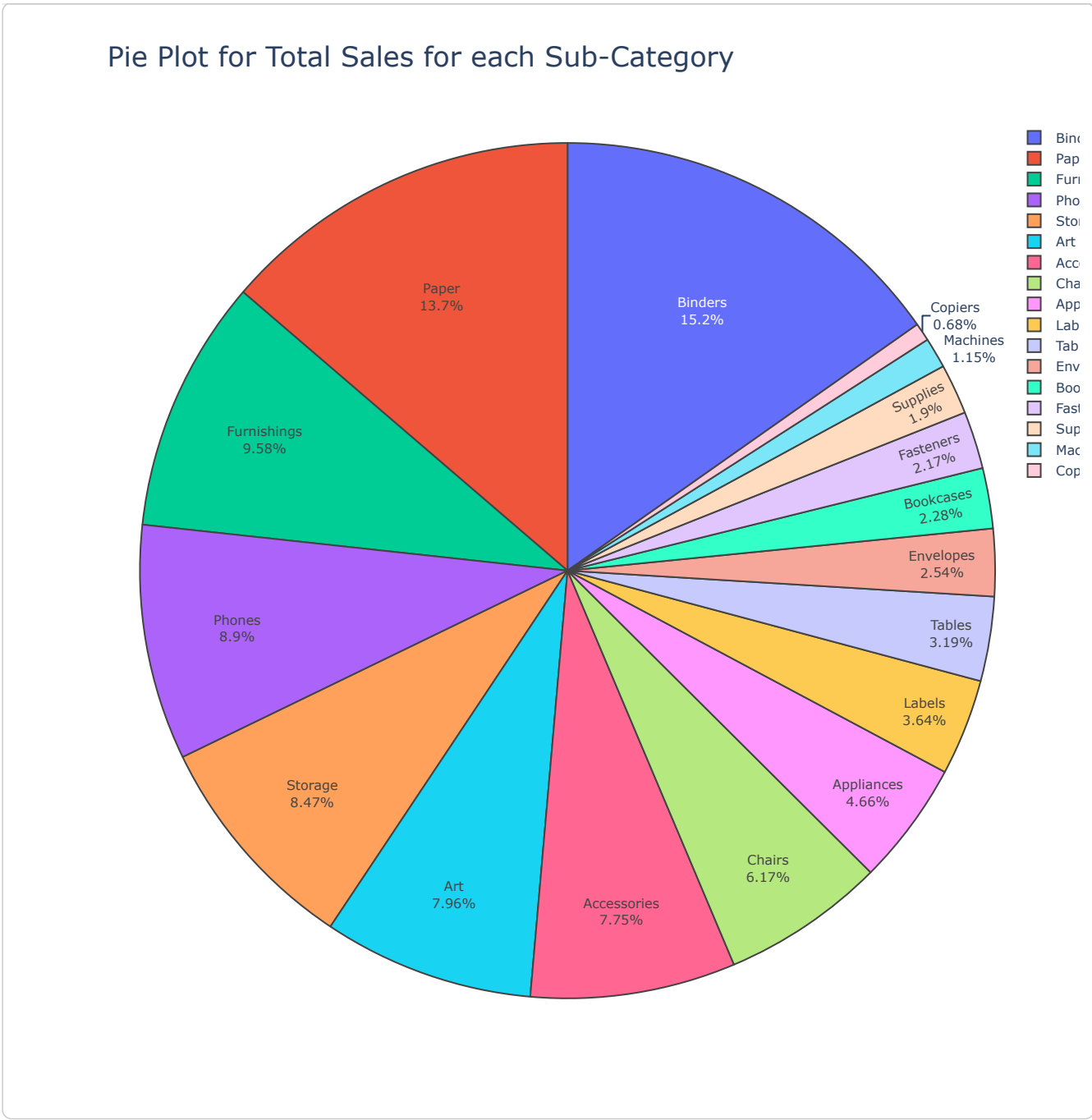


Category → Sub-Category (Sunburst Chart)

The sunburst chart gives a hierarchical breakdown: Inner ring → Category Outer ring → Sub-Category The size of each segment represents Sales contribution. Hovering shows detailed metrics like Sales, Quantity, and Profit.

Insight: One can identify Which sub-categories drive the most sales Which ones contribute less (or may need improvement) It reveals hidden patterns not visible in the pie chart (like which sub-category dominates inside a category)

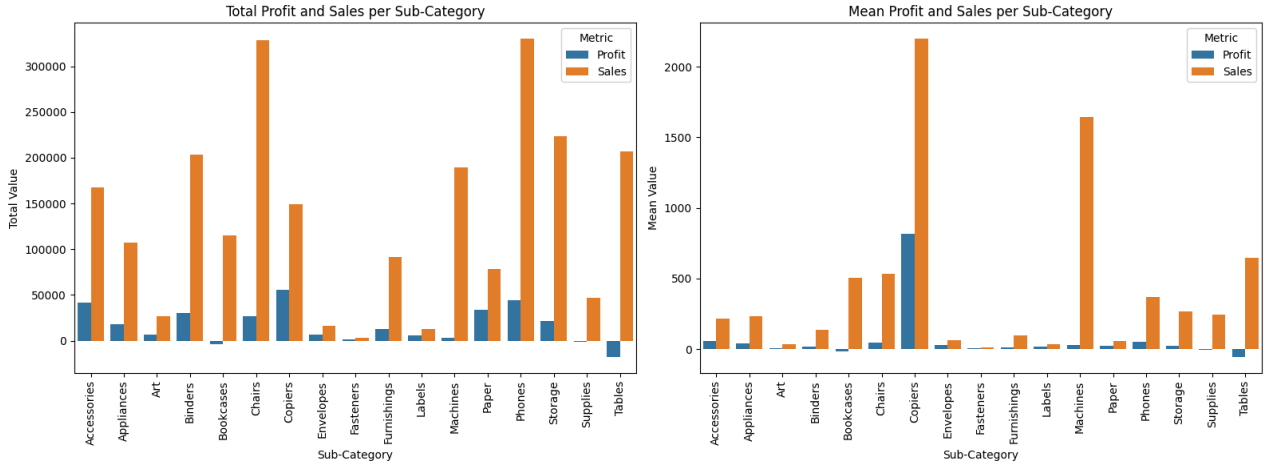
```
import plotly.graph_objects as go
from plotly.offline import iplot

data = go.Pie(
    labels=df['Sub-Category'].value_counts().index,
    values=df['Sub-Category'].value_counts(),
    textinfo='label+percent',
    hoverinfo='label+percent',
    marker=dict(line=dict(width=1.5))
)

layout = go.Layout(
    title='Pie Plot for Total Sales for each Sub-Category',
    titlefont=dict(size=25),
    height=1000
)

iplot(dict(data=[data], layout=layout))
```



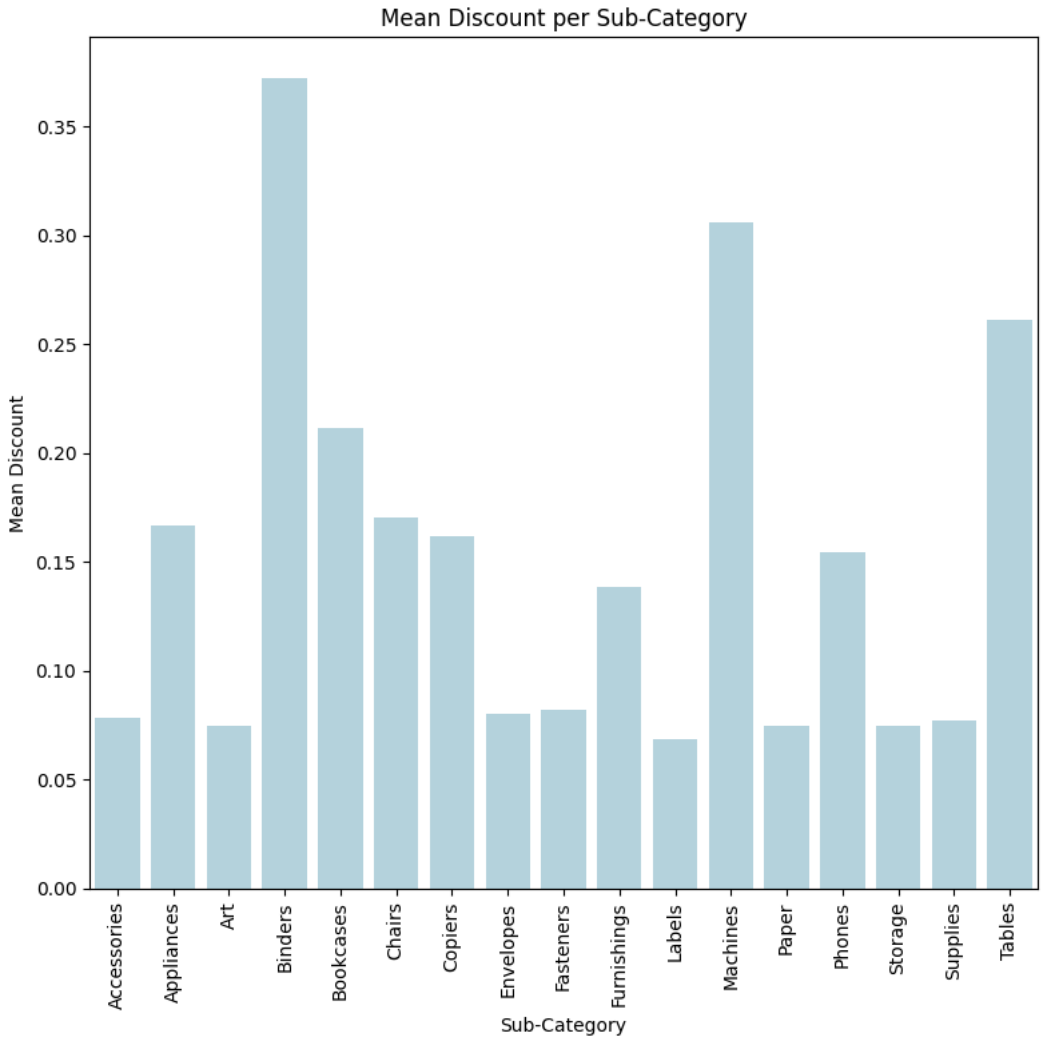
Review of sales and profits

```
# Grouping
grouped_sum = df.groupby('Sub-Category')[['Profit', 'Sales']].sum().reset_index()
grouped_mean = df.groupby('Sub-Category')[['Profit', 'Sales']].mean().reset_index()

# Create subplots
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Total values plot
sns.barplot(
    data=grouped_sum.melt(id_vars='Sub-Category', var_name='Metric', value_name='Value'),
    x='Sub-Category',
    y='Value',
    hue='Metric',
    ax=axes[0]
)
axes[0].set_title('Total Profit and Sales per Sub-Category')
axes[0].set_ylabel('Total Value')
axes[0].tick_params(axis='x', rotation=90)

# Mean values plot
sns.barplot(
    data=grouped_mean.melt(id_vars='Sub-Category', var_name='Metric', value_name='Value'),
    x='Sub-Category',
    y='Value',
    hue='Metric',
```

```
    ax=axes[1]
)
axes[1].set_title('Mean Profit and Sales per Sub-Category')
axes[1].set_ylabel('Mean Value')
axes[1].tick_params(axis='x', rotation=90)

plt.tight_layout()
plt.show()
```



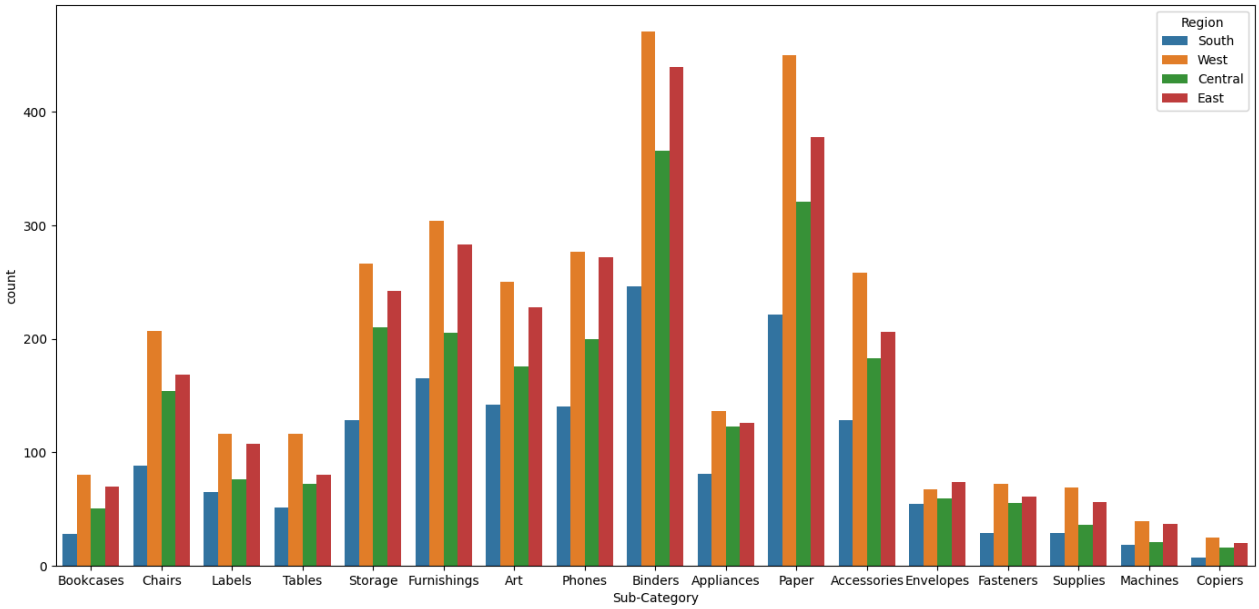

```
grouped_discount_mean = df.groupby('Sub-Category')['Discount'].mean().reset_index()

fig, ax = plt.subplots(figsize=(8, 8))

sns.barplot(
    data=grouped_discount_mean,
    x='Sub-Category',
    y='Discount',
    ax=ax,
    color='lightblue'
)

ax.set_title('Mean Discount per Sub-Category')
ax.set_ylabel('Mean Discount')
ax.tick_params(axis='x', rotation=90)

plt.tight_layout()
plt.show()
```



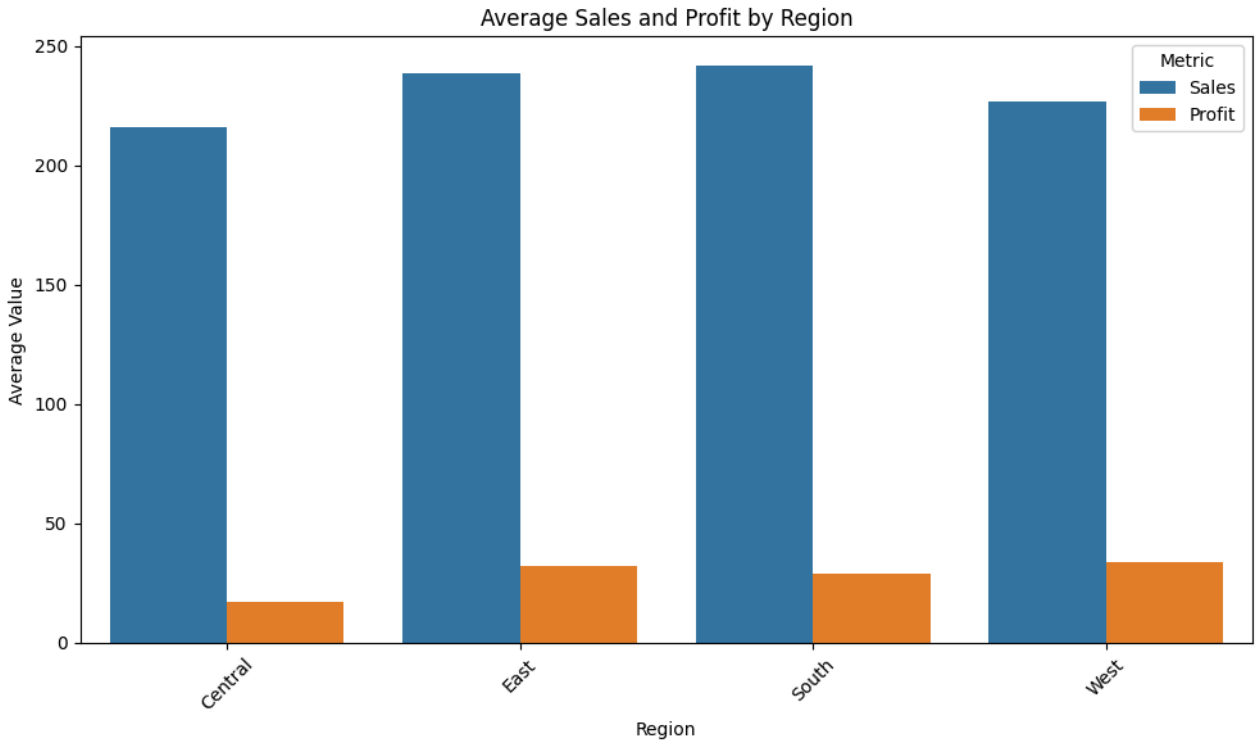
;

Review of the sales of different parts of the US

```
plt.figure(figsize=(17,8))
sns.countplot(x="Sub-Category", hue="Region", data=df)
plt.show()
```



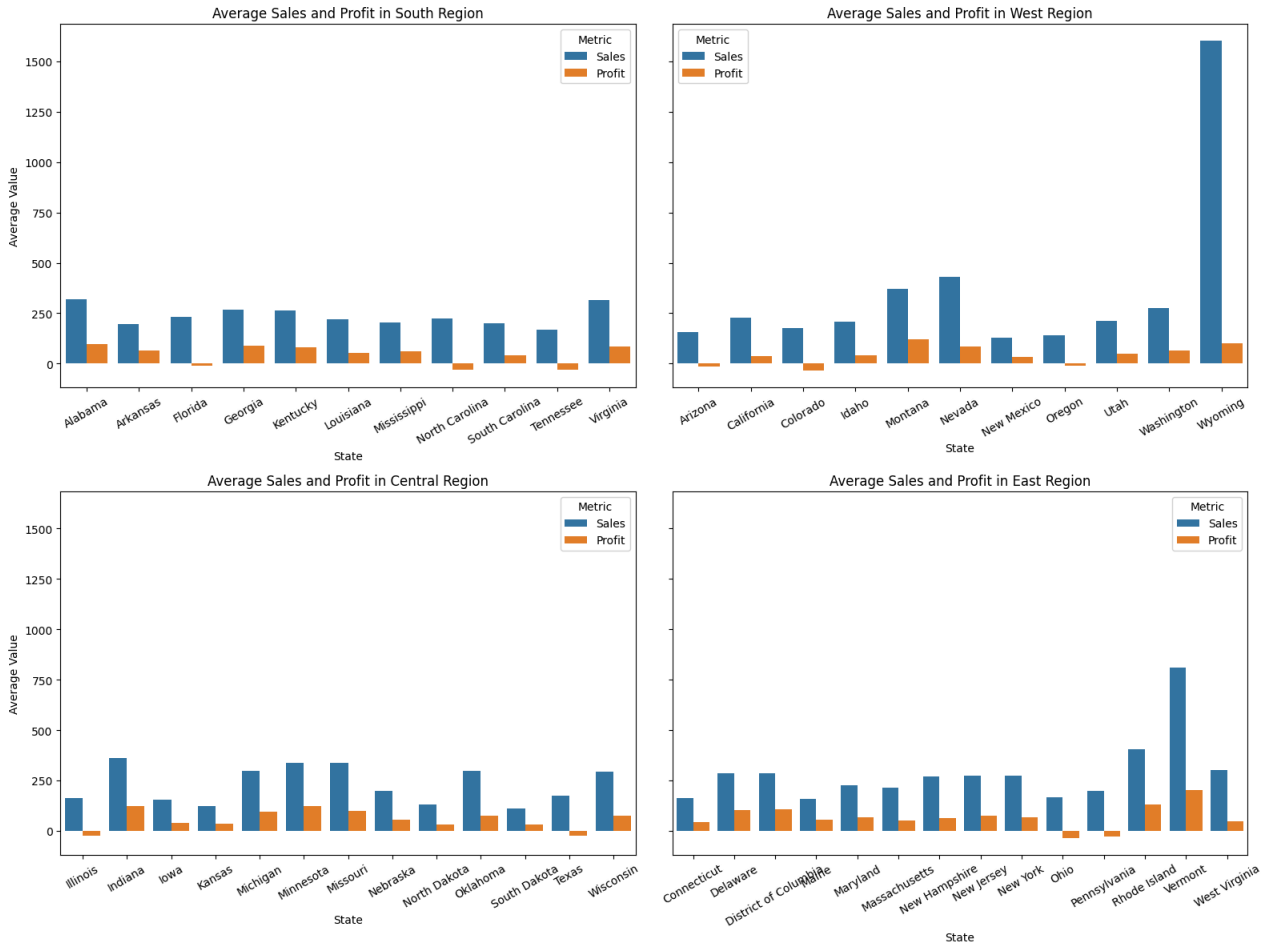
```
grouped_data = df.groupby('Region')[['Sales', 'Profit']].mean().reset_index()

melted_data = grouped_data.melt(
    id_vars='Region',
    var_name='Metric',
    value_name='Value'
)

plt.figure(figsize=(10, 6))

sns.barplot(
    data=melted_data,
    x='Region',
    y='Value',
    hue='Metric'
)

plt.title('Average Sales and Profit by Region')
plt.ylabel('Average Value')
plt.xlabel('Region')

plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```



|||

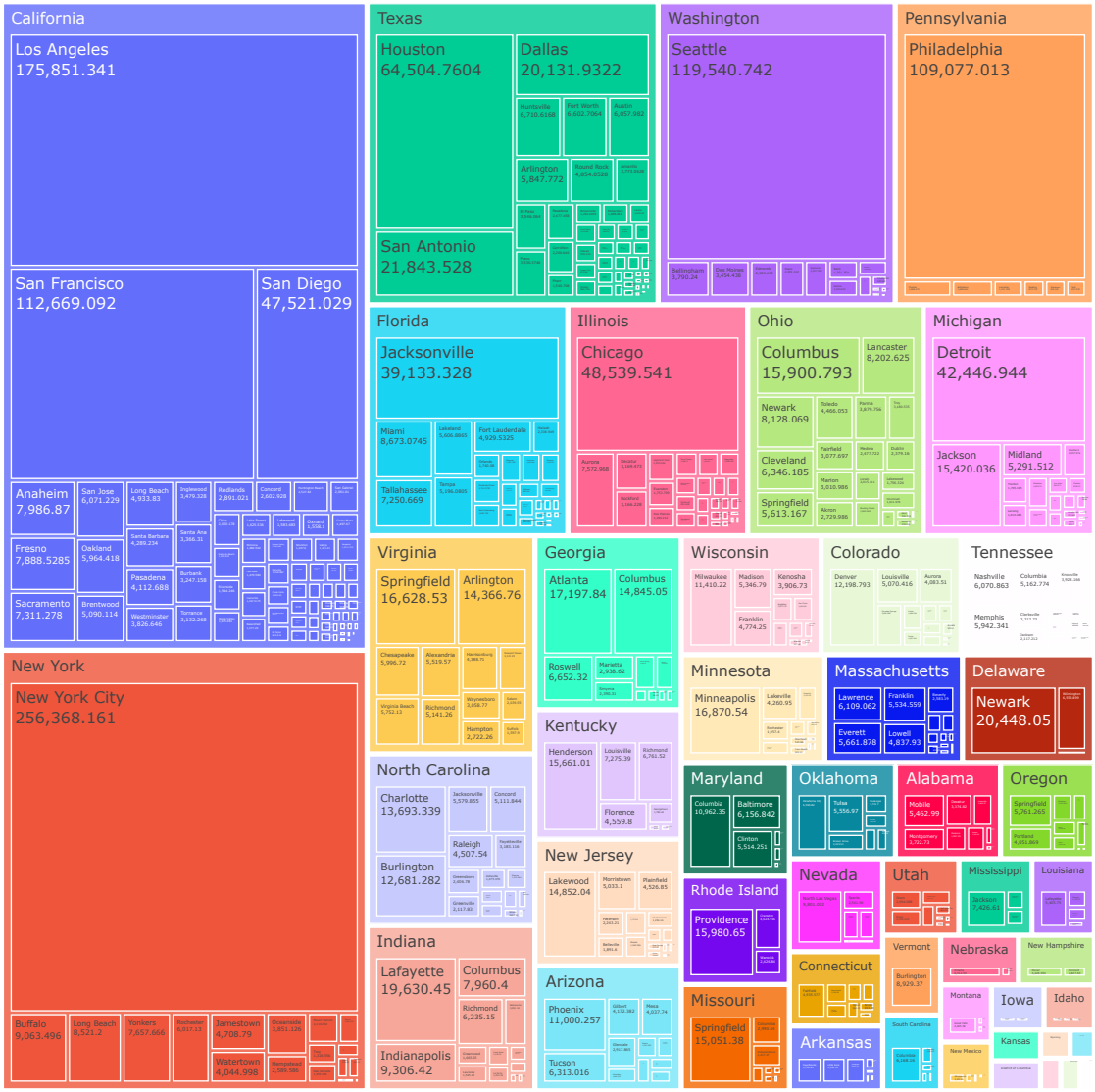
```
regions = df['Region'].unique()

fig, axes = plt.subplots(2, 2, figsize=(16, 12), sharey=True)
axes = axes.flatten()

for i in range(4):
    df_region = df[df['Region'] == regions[i]].copy()

    grouped_data = df_region.groupby('State')[['Sales', 'Profit']].mean().reset_index()

    melted_data = grouped_data.melt(
        id_vars='State',
        var_name='Metric',
        value_name='Value'
    )

    sns.barplot(
        data=melted_data,
        x='State',
        y='Value',
        hue='Metric',
```

```
        ax=axes[i]
    )

    axes[i].set_title(f'Average Sales and Profit in {regions[i]} Region')
    axes[i].set_ylabel('Average Value')
    axes[i].tick_params(axis='x', rotation=30)
    axes[i].legend(title='Metric')

plt.tight_layout()
plt.show()
```



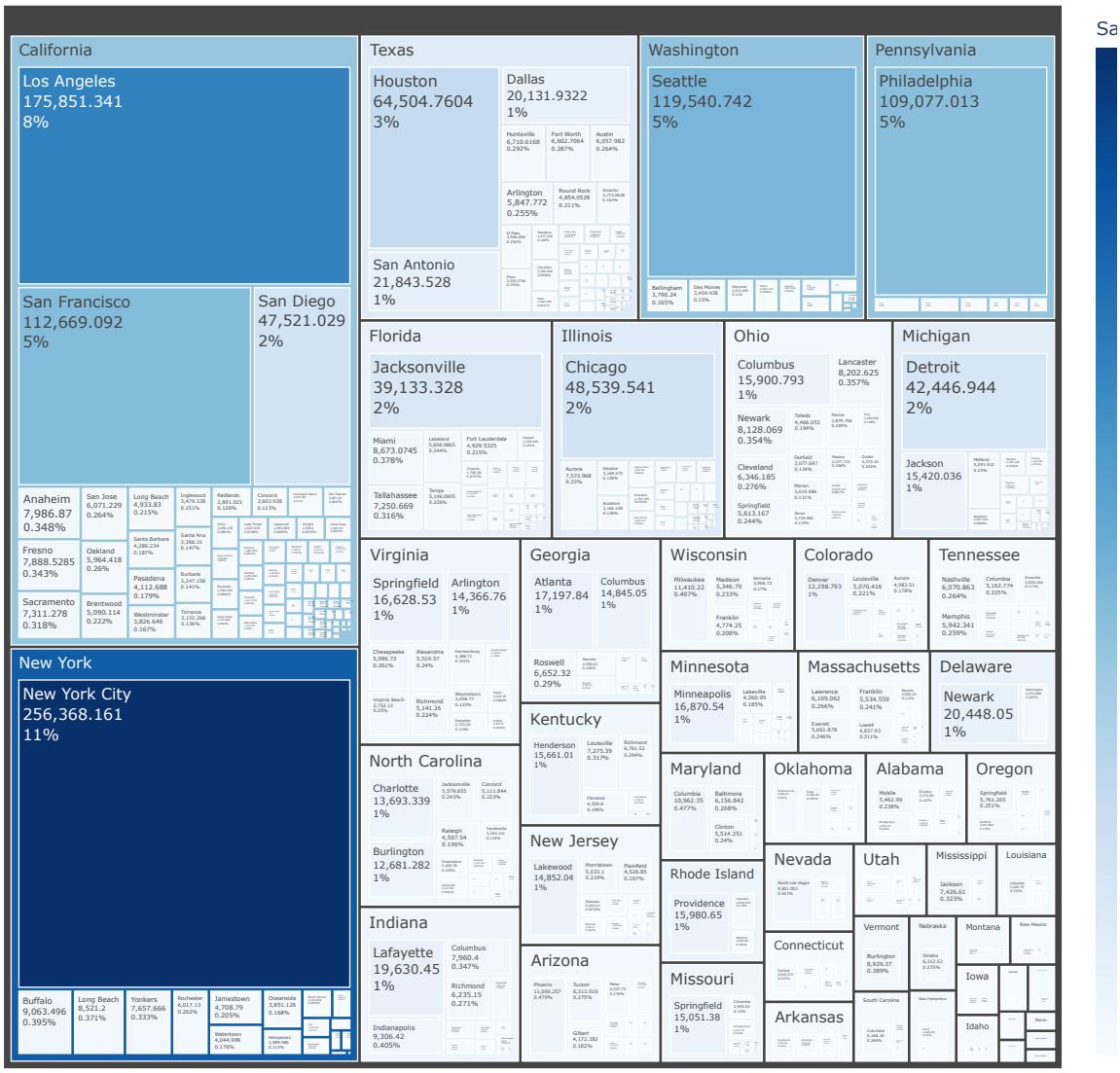

```
import plotly.express as px

temp = df[['State', 'City', 'Sales']] \
    .groupby(['State', 'City'])['Sales'] \
    .sum() \
    .reset_index()

fig = px.treemap(
    temp,
    path=['State', 'City'],
    values='Sales'
)

fig.update_layout(
    height=1000,
    title='City-wise Sales'

)

fig.data[0].textinfo = 'label+text+value'

fig.show()
```

City-wise Sales




```
temp = df[['State', 'City', 'Sales']] \
    .groupby(['State', 'City'])['Sales'] \
    .sum() \
    .reset_index()

fig = px.treemap(
    temp,
    path=['State', 'City'],
    values='Sales',
    color='Sales',
    color_continuous_scale='Blues',
    title='City-wise Sales Distribution'
)

fig.update_layout(
    height=1000
)


fig.data[0].textinfo = 'label+value+percent entry'

fig.update_traces(
    hovertemplate='<b>%{label}</b><br>Sales: %{value}<br>Share: %{percentEntry:.2%}<extra></extra>'
)

fig.show()
```

City-wise Sales Distribution



uuuu

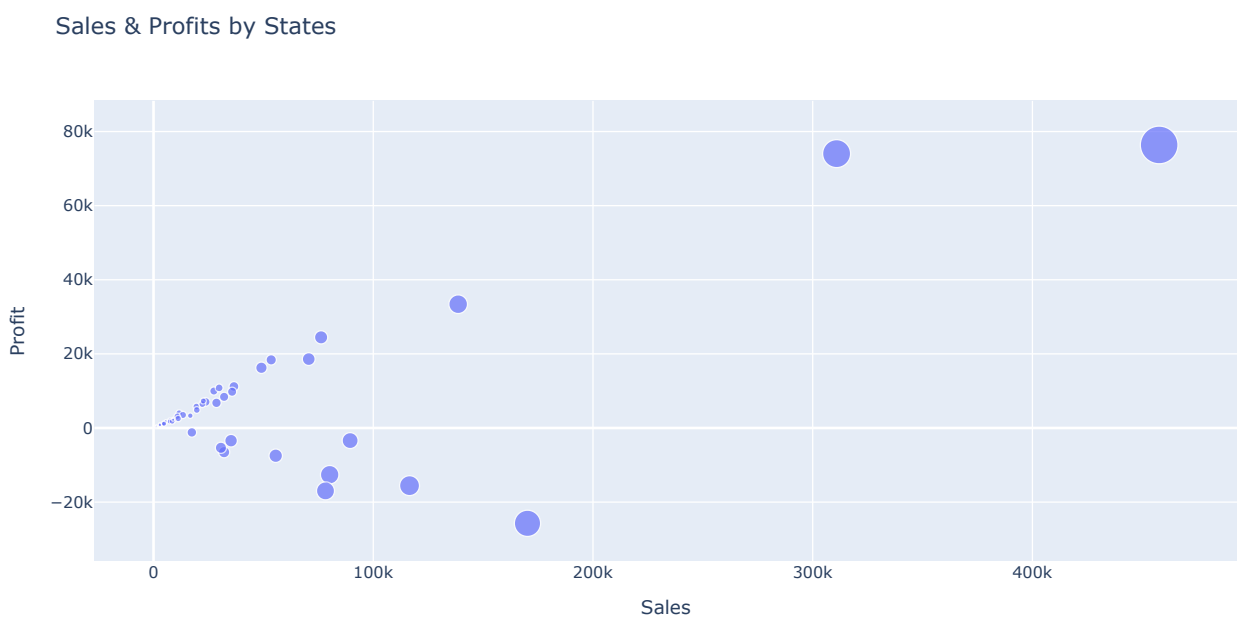
```
df_notime = df[['Segment', 'City', 'State', 'Region', 'Category', 'Sub-Category', 'Sales', 'Quantity', 'Discount', 'Profit
```

```
df
```

	Order Date	Segment	City	State	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
0	2016-11-08	Consumer	Henderson	Kentucky	South	Furniture	Bookcases	261.9600	2	0.00	41.9136
1	2016-11-08	Consumer	Henderson	Kentucky	South	Furniture	Chairs	731.9400	3	0.00	219.5820
2	2016-06-12	Corporate	Los Angeles	California	West	Office Supplies	Labels	14.6200	2	0.00	6.8714
3	2015-10-11	Consumer	Fort Lauderdale	Florida	South	Furniture	Tables	957.5775	5	0.45	-383.0310
4	2015-10-11	Consumer	Fort Lauderdale	Florida	South	Office Supplies	Storage	22.3680	2	0.20	2.5164
...	...	...	...	...	...	...	...	...	...	...	...
9989	2014-01-21	Consumer	Miami	Florida	South	Furniture	Furnishings	25.2480	3	0.20	4.1028
9990	2017-02-26	Consumer	Costa Mesa	California	West	Furniture	Furnishings	91.9600	2	0.00	15.6332
9991	2017-02-26	Consumer	Costa Mesa	California	West	Technology	Phones	258.5760	2	0.20	19.3932
9992	2017-02-26	Consumer	Costa Mesa	California	West	Office Supplies	Paper	29.6000	4	0.00	13.3200

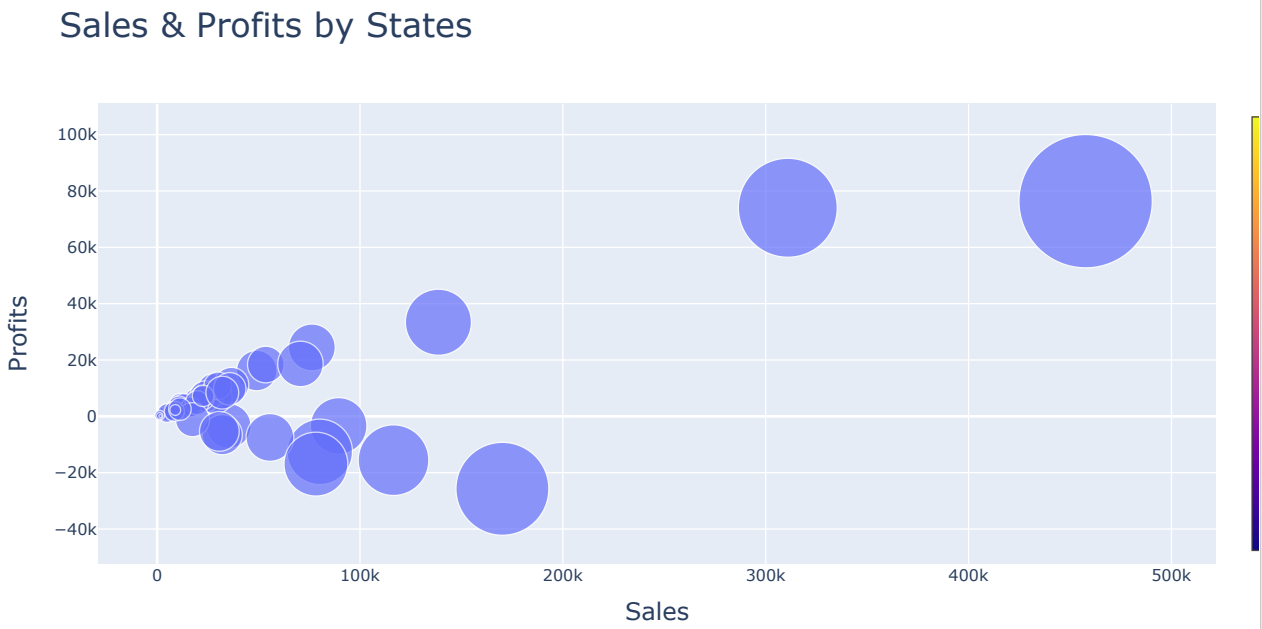
Next steps: Generate code with df New interactive sheet

```
state_data = df_notime.groupby('State').sum().reset_index()

fig = px.scatter(
    state_data,
    x='Sales',
    y='Profit',
    size='Quantity',
    hover_name='State',
    title='Sales & Profits by States'
)

fig.show()
```




```
#Advanced Analysis

state_data = df_notime.groupby('State').sum().reset_index()

fig = go.Figure(
    data=go.Scatter(
        x=state_data['Sales'],
        y=state_data['Profit'],
        mode='markers',
```

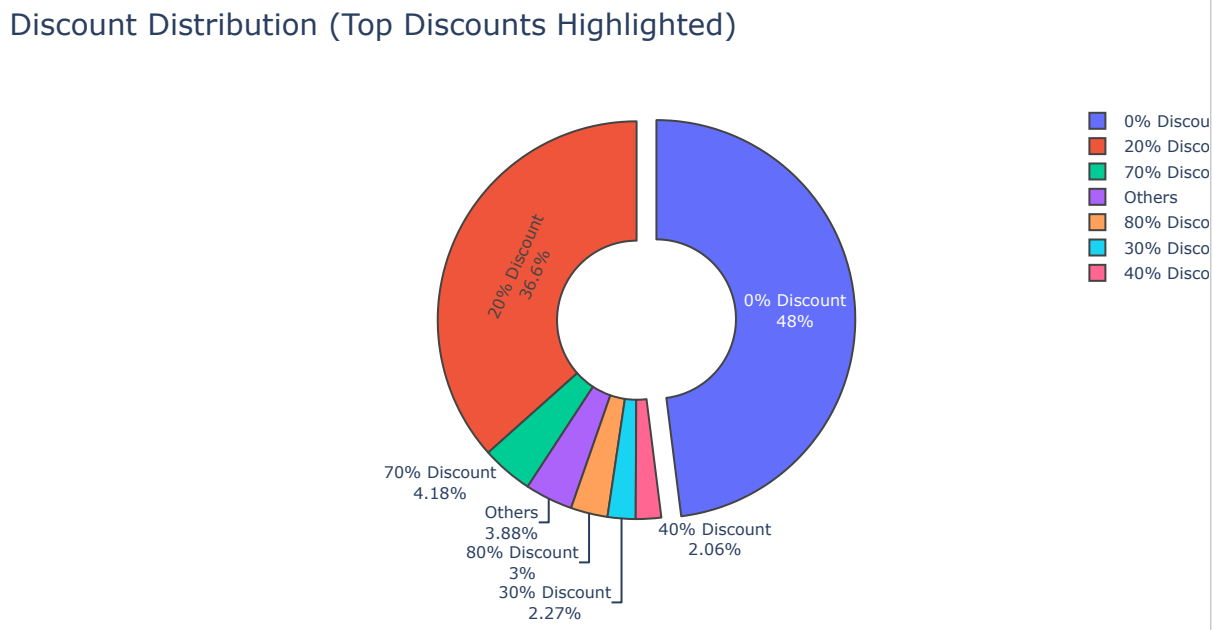
```
        text=state_data['State'],
        marker=dict(
            size=state_data['Quantity'],
            sizemode='area',
            sizeref=2.*max(state_data['Quantity'])/(100.**2),
            showscale=True
        )
    )
)

fig.update_layout(
    title='Sales & Profits by States',
    title_font=dict(size=25),
    xaxis=dict(title='Sales', title_font=dict(size=18)),
    yaxis=dict(title='Profits', title_font=dict(size=18))
)

fig.show()
```



Discount Analysis

```
discount_counts = df_notime['Discount'].value_counts().sort_values(ascending=False)


top_n = 6
top_discounts = discount_counts.head(top_n)
others = discount_counts.iloc[top_n:].sum()


values = list(top_discounts.values)
labels = [f"{int(d*100)}% Discount" for d in top_discounts.index]

if others > 0:
    values.append(others)
    labels.append('Others')


fig = go.Figure(
    data=[go.Pie(
        labels=labels,
        values=values,
        hole=0.4,
        textinfo='label+percent',
        hoverinfo='label+percent+value',
        marker=dict(line=dict(width=1.5)),
        pull=[0.1 if i == 0 else 0 for i in range(len(labels))]
    )]
)

fig.update_layout(
    title='Discount Distribution (Top Discounts Highlighted)',
    title_font=dict(size=22)
```

```
)

fig.show()
```




```
discount_counts = df_notime['Discount'].value_counts().sort_index()
```

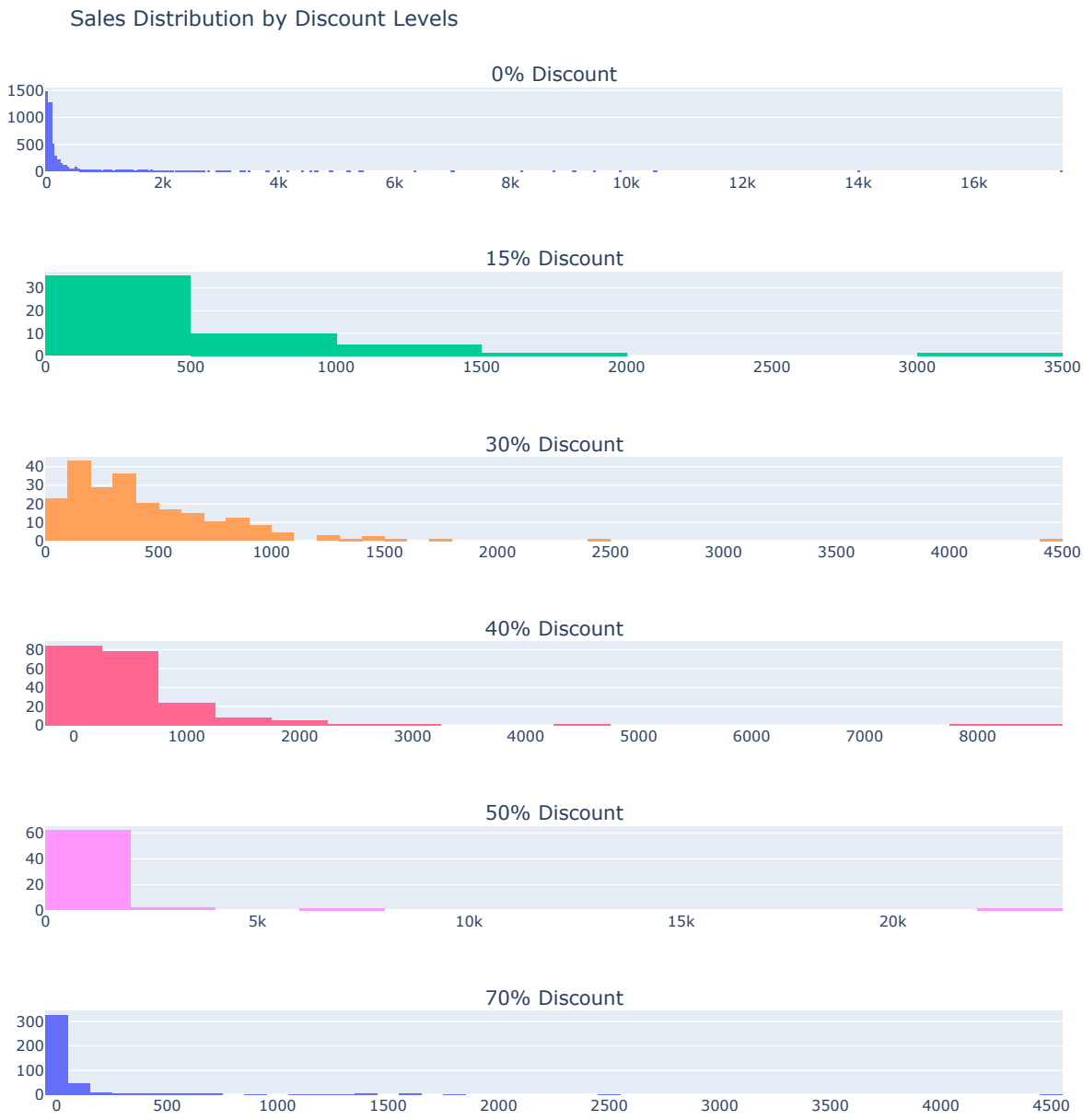
A few discount levels dominate most transactions

Remaining discount levels contribute relatively less and are grouped as “Others”

Helps identify most frequently used pricing strategies

```
import plotly.graph_objects as go

data = [
    go.Histogram(
        x=df_notime.sort_values('Discount')['Discount']
    )
]

layout = go.Layout(
    title='Sales Histogram Plot For Each Discount',
    xaxis=dict(
        title='Discount',
        title_font=dict(size=20)
    )
)

fig = dict(data=data, layout=layout)

iplot(fig)
```

```
from plotly.subplots import make_subplots

fig = make_subplots(
    rows=6,
    cols=2,
    subplot_titles=[
        '0% Discount','10% Discount','15% Discount',
        '20% Discount','30% Discount','32% Discount',
        '40% Discount','45% Discount','50% Discount',
        '60% Discount','70% Discount','80% Discount'
    ],
    horizontal_spacing=0.1,
    vertical_spacing=0.1
)

fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0]['Sales'], name='0% Distribution'), row=1, col=1)
fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.1]['Sales'], name='10% Distribution'), row=1, col=2)

fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.15]['Sales'], name='15% Distribution'), row=2, col=1)
fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.2]['Sales'], name='20% Distribution'), row=2, col=2)

fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.3]['Sales'], name='30% Distribution'), row=3, col=1)
fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.32]['Sales'], name='32% Distribution'), row=3, col=2)

fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.4]['Sales'], name='40% Distribution'), row=4, col=1)
fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.45]['Sales'], name='45% Distribution'), row=4, col=2)

fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.5]['Sales'], name='50% Distribution'), row=5, col=1)
fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.6]['Sales'], name='60% Distribution'), row=5, col=2)

fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.7]['Sales'], name='70% Distribution'), row=6, col=1)
fig.add_trace(go.Histogram(x=df_notime[df_notime['Discount']==0.8]['Sales'], name='80% Distribution'), row=6, col=2)

fig.update_layout(
    width=2000,
    height=1000,
    title_text='Sales Distribution by Discount Levels',
    showlegend=False
)

fig.show()
```



By applying the right policy on discounts,the loss of the store can be avoided in some sections

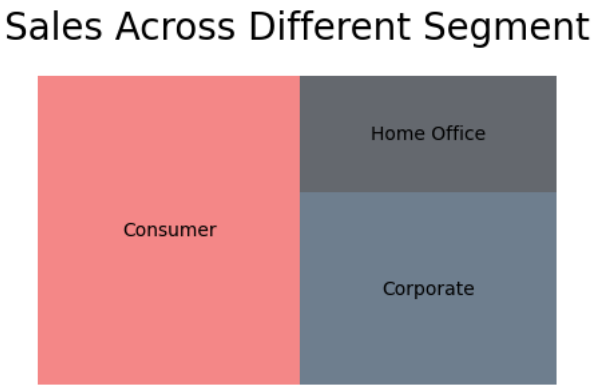
SEGMENTATION

```
!pip install squarify
```

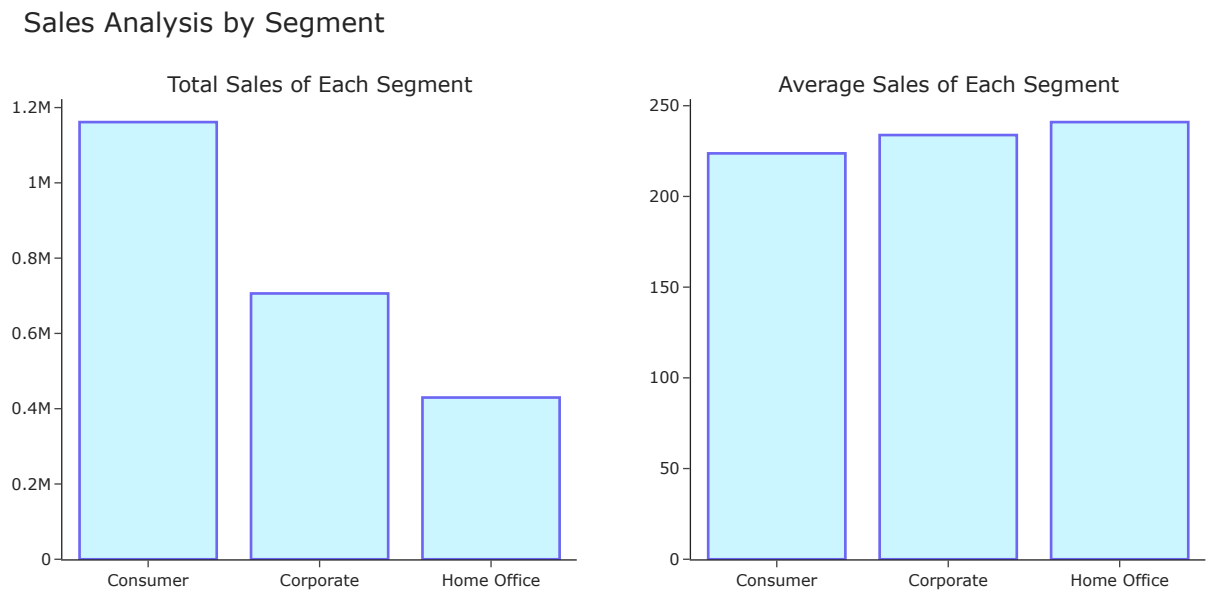
```
Requirement already satisfied: squarify in /usr/local/lib/python3.12/dist-packages (0.4.4)
```

```
import squarify

df_tree = df[['Segment', 'Sales']]
df_tree = df_tree.groupby(['Segment']).sum().reset_index()

plt.figure(figsize=(5, 3))

squarify.plot(
    sizes=df_tree['Sales'],
    label=df_tree['Segment'],
    color=['#F05454', '#30475E', '#222831'],
    alpha=0.7
)

plt.title("Sales Across Different Segment", size=20, pad=20)
```

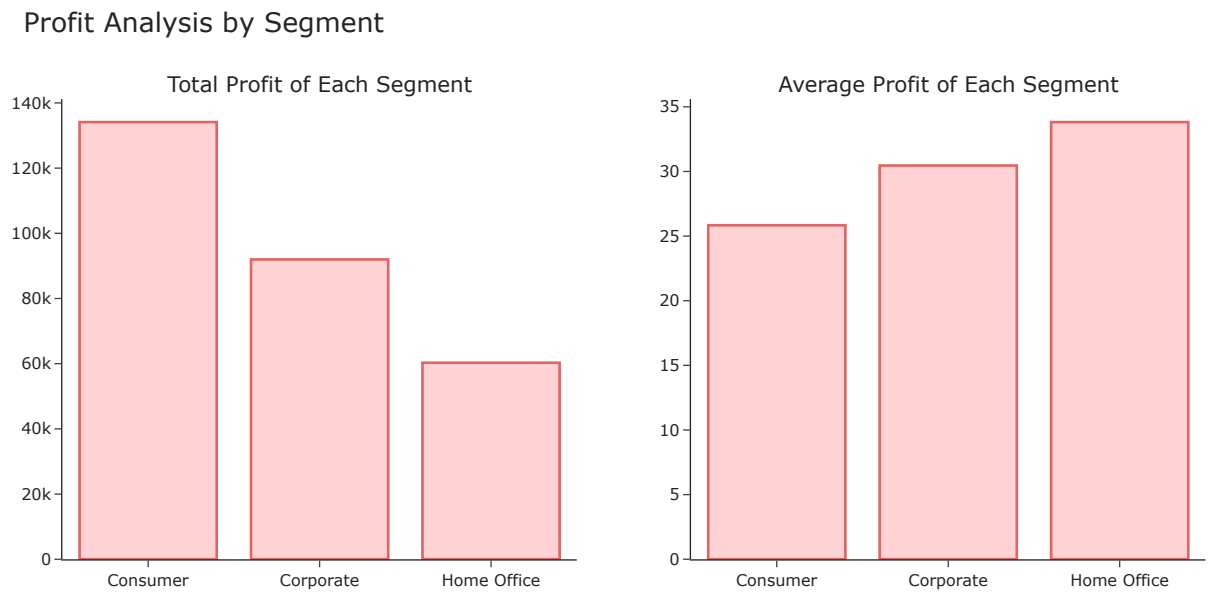
```
plt.axis('off')

plt.show()
```



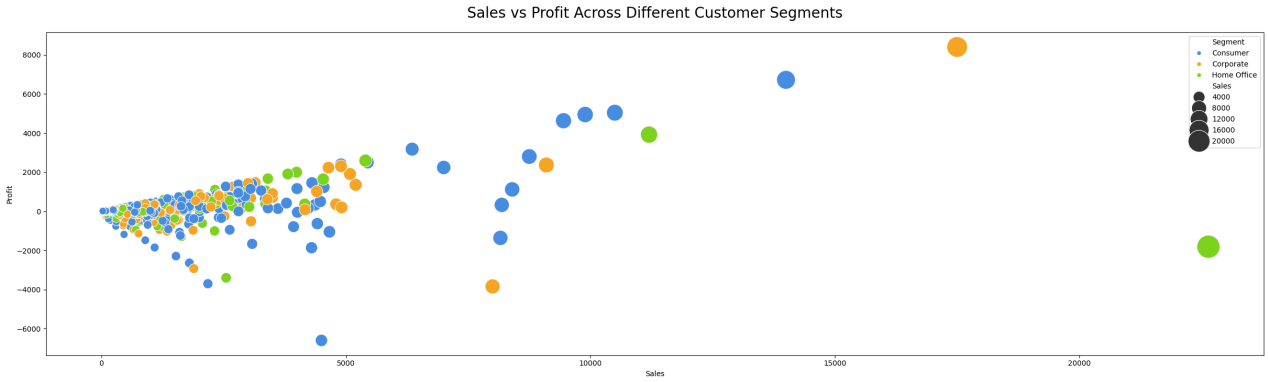

```
from plotly.subplots import make_subplots
import plotly.graph_objects as go


segments = df['Segment'].unique()


total_sales = [df[df['Segment'] == i]['Sales'].sum() for i in segments]
average_sales = [df[df['Segment'] == i]['Sales'].mean() for i in segments]


fig = make_subplots(
    rows=1,
    cols=2,
    subplot_titles=[
        'Total Sales of Each Segment',
        'Average Sales of Each Segment'
    ]
)


fig.add_trace(
    go.Bar(
        x=segments,
        y=total_sales,
        marker_color='rgb(171,241,255)',
        marker_line_color='rgb(12,0,235)',
        marker_line_width=2,
        opacity=0.6
    ),
    row=1, col=1
)

fig.add_trace(
    go.Bar(
        x=segments,
        y=average_sales,
        marker_color='rgb(171,241,255)',
        marker_line_color='rgb(12,0,235)',
        marker_line_width=2,
        opacity=0.6
    ),
    row=1, col=2
)


fig.update_layout(
    template='simple_white',
    title='Sales Analysis by Segment',
    title_font_size=20,
    showlegend=False
)

fig.show()
```




```
total_profit = [df[df['Segment'] == i]['Profit'].sum() for i in segments]
average_profit = [df[df['Segment'] == i]['Profit'].mean() for i in segments]


fig = make_subplots(
    rows=1,
    cols=2,
    subplot_titles=[
        'Total Profit of Each Segment',
        'Average Profit of Each Segment'
    ]
)

fig.add_trace(
    go.Bar(
        x=segments,
        y=total_profit,
        marker_color='rgb(255,183,183)',
        marker_line_color='rgb(200,0,0)',
        marker_line_width=2,
        opacity=0.6
    ),
    row=1, col=1
)

fig.add_trace(
    go.Bar(
        x=segments,
        y=average_profit,
        marker_color='rgb(255,183,183)',
        marker_line_color='rgb(200,0,0)',
        marker_line_width=2,
        opacity=0.6
    ),
    row=1, col=2
)

fig.update_layout(
    template='simple_white',
    title='Profit Analysis by Segment',
    title_font_size=20,
    showlegend=False
)

fig.show()
```




```
df_scatter = df[['Sales', 'Profit', 'Segment']].copy()


plt.figure(figsize=(30, 8))

sns.scatterplot(
    x=df_scatter['Sales'],
    y=df_scatter['Profit'],
    hue=df_scatter['Segment'],
    palette=['#4A90E2', '#F5A623', '#7ED321'],
    size=df_scatter['Sales'],
    sizes=(100, 1000),
    legend='auto'
)

plt.title("Sales vs Profit Across Different Customer Segments", size=20, pad=20)

plt.show()
```



sllla

Analyze the Orders & Sales

```
df_grouped = df.groupby('Segment').agg({
    'Sales': 'sum',
    'Profit': ['sum', 'mean']
}).reset_index()

df_grouped.columns = ['Segment', 'Total Sales', 'Total Profit', 'Avg Profit']


fig = make_subplots(
```

```
    rows=2, cols=2,
    specs=[[{"type": "bar"}, {"type": "bar"}],
           [{"type": "scatter"}, {"type": "domain"}]],
    subplot_titles=[
        "Total Sales by Segment",
        "Total Profit by Segment",
        "Sales vs Profit",
        "Segment Contribution (Treemap)"
    ]
)


fig.add_trace(
    go.Bar(
        x=df_grouped['Segment'],
        y=df_grouped['Total Sales'],
        marker=dict(color='skyblue'),
        text=df_grouped['Total Sales'],
        textposition='outside'
    ),
    row=1, col=1
)

fig.add_trace(
    go.Bar(
        x=df_grouped['Segment'],
        y=df_grouped['Total Profit'],
        marker=dict(color='salmon'),
        text=df_grouped['Total Profit'],
        textposition='outside'
    ),
    row=1, col=2
)


fig.add_trace(
    go.Scatter(
        x=df['Sales'],
        y=df['Profit'],
        mode='markers',
        marker=dict(
            size=df['Sales']/50,
            color=df['Profit'],
            colorscale='Viridis',
            showscale=True
        ),
        text=df['Segment']
    ),
    row=2, col=1
)


treemap = px.treemap(
    df,
    path=['Segment'],
    values='Sales'
)

fig.add_trace(
    treemap.data[0],
    row=2, col=2
)

fig.update_layout(
    height=900,
    width=1200,
    title_text="Sales & Profit Dashboard",
    template="plotly_white",
    showlegend=False
)

fig.show()
```

```
df_time = df[['Order Date', 'Segment', 'Sales', 'Quantity', 'Discount', 'Profit']].copy()

df_time['Month_Name'] = df_time['Order Date'].dt.month_name()
df_time['Year_Month'] = df_time['Order Date'].dt.to_period('M')
```

```
df_line = df_time[['Order Date', 'Sales', 'Profit']].sort_values('Order Date')
```